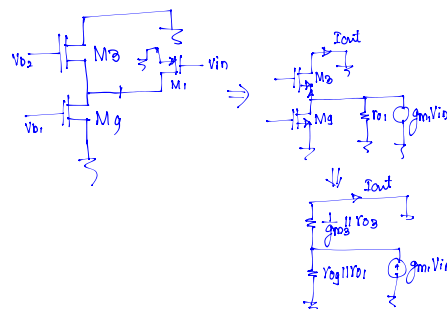
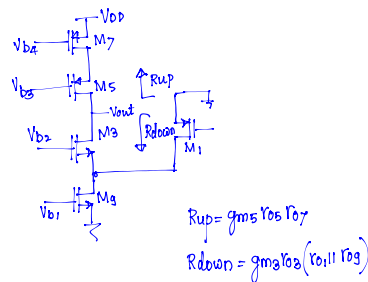
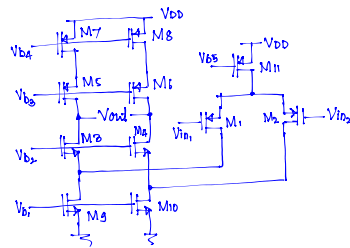
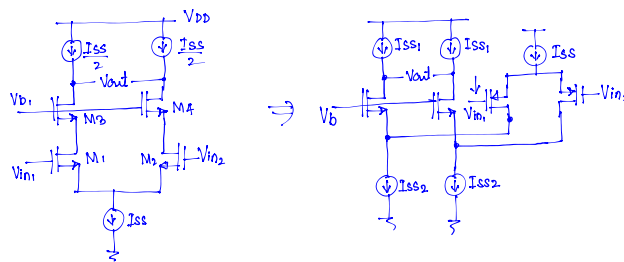
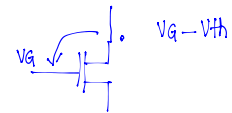
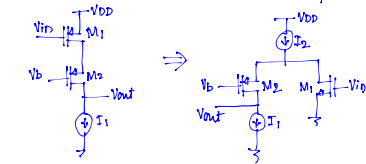
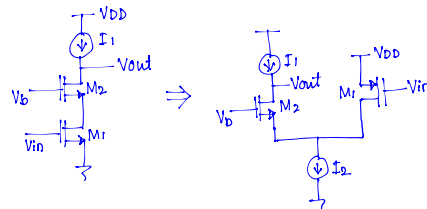


Folded cascode op-amps:-



$$I_{out} = g_{m1} V_{in} \times \frac{r_{o1} || r_{o9}}{\frac{1}{g_{m3}} + (r_{o3} + (r_{o1} || r_{o9}))}$$

$$I_{out} \approx g_{m1} V_{in} \times \frac{r_{o1} || r_{o9}}{r_{o3} || r_{o9}}$$

$$g_m = \frac{I_{out}}{V_{in}} = g_{m1}$$